Exercise Sheet 2
Date of discussion: 05. November 2026
Supervisor: Emiliia Gelóczi

Question 1: Cryptography and the Adversary

Alice and Bob communicate over an insecure channel.

a) Explain the difference between cryptography, cryptanalysis, and cryptology.

b) Compare the old and new paradigms of cryptography. Why might a trusted third party be necessary?

c) Explain unconditional and conditional security. What is the fundamental difference?

d) Alice encrypts her messages. Can an attacker still delete or block them? Which security property would be affected?

Question 2: Cryptographic Hash Functions

a) Explain the following two fundamental properties of hash functions in your own words:

- Ease of computation
- Compression

b) Define the following security properties of cryptographic hash functions using mathematical notation. Explain the differences between them.

- Pre-image resistance
- Second pre-image resistance
- Collision resistance

c) Consider a cryptographic hash function h. Match each of the following situations to the corresponding security property from part (b).

a) Given a hash value y, find a message m such that $h(m) = y$.

b) Given a message m, find another message $m' \neq m$ such that $h(m) = h(m')$.

c) Find any two different messages $m \neq m'$ such that $h(m) = h(m')$.

d) Consider an ideal cryptographic hash function h that produces a hash value of $n = 32$ bits. Assume that hash values are uniformly distributed and independent for randomly selected distinct inputs. How many different hash values can the function produce?

e) Explain the difference between a Manipulation Detection Code (MDC) and a Message Authentication Code (MAC). Why is an ordinary cryptographic hash value alone insufficient to authenticate the sender of a message?